1 Introduction and inequalities

Week 1 · Week summary

Core ideas, lesson by lesson

1 The real numbers and the number line

- $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, and the irrationals complete $\mathbb{R}$.
- Every real number is one point on the line; left means smaller.
- $|a - b|$ is the distance between a and b.

2 Equations, degree, and roots

- Move everything to one side: $P(x) = 0$.
- The degree is the highest power after simplifying.
- Degree n means at most n real roots; check roots by substitution.

3 Linear and quadratic equations

- Linear: one root, $x = -b/a$.
- Quadratic: factor, isolate x^2, or use the formula.
- $\Delta = b^2 - 4ac$ decides 2, 1, or 0 real roots; in word problems discard roots that make no sense.

4 Higher-degree and special equations

- Rational root candidates: ± (divisors of the constant) / (divisors of the leading coefficient).
- Remainder 0 in synthetic division means you found a root.
- After squaring or clearing denominators, check every answer.

5 Inequalities and interval notation

- [ ] includes the endpoint, ( ) excludes it; ∞ always takes ( ).
- Filled dot = included, hollow dot = excluded.
- "or" is a union ∪; "and" is an intersection ∩.

6 Properties and linear inequalities

- Add, subtract, or multiply by a positive: the sign stays.
- Multiply or divide by a negative: flip the sign.
- Check one value from your answer in the original inequality.

7 Polynomial and rational inequalities

- Zero on one side, factor, find critical points, test one value per interval.
- ≤ or ≥ include the zeros of the numerator; < or > exclude them.
- Zeros of a denominator are always excluded.

8 Absolute value and systems of inequalities

- $|x - a| < r \rightarrow a - r < x < a + r$ (one interval).
- $|x - a| > r \rightarrow x < a - r$ or $x > a + r$ (a union).
- A system keeps the intersection of the solutions.

Interval	**x < -4**	**-4 < x < 3**	**x > 3**
Test value	x = -5	x = 0	x = 4
x + 4	**-**	**+**	**+**
x - 3	**-**	**-**	**+**
(x + 4)(x - 3)	**+**	**-**	**+**

> 0



Watch out

Mistake	Fix
Forgetting to flip the sign when dividing by a negative number.	Circle every negative divisor before dividing.
Including a zero of the denominator in the answer.	Mark those points with a hollow dot from the start.
Thinking $-5 > -2$ because $5 > 2$.	Picture the line: -5 is further left, so $-5 < -2$.
Writing $x^2 = 49 \rightarrow x = 7$ and forgetting -7.	Taking a square root gives $\pm$: $x = 7$ or $x = -7$.
Dividing both sides by x and losing the root $x = 0$.	Factor x out instead: $x(x - 3) = 0$ keeps both roots.
Keeping a root that appeared only after squaring.	Substitute each candidate into the original equation.
Writing $[3, \infty]$.	∞ is never included: $[3, \infty)$.
Flipping the sign when subtracting a negative number.	Only multiplying or dividing by a negative flips it.